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	Sultanate of Oman Ministry of Health The Royal Hospital Department of Blood Transfusion Services

Code: BTS-GUD-3-Vers.2

Effective Date: 24/09/2023
Target Review Date: 31/08/2025

Title: Irradiated blood components.

1. Introduction:

Irradiated blood components are required for patients at risk for developing transfusion graft versus host disease (TR-GVHD) which is fatal and yet preventable. The risk associated with an individual transfusion depends on the number and viability of contaminating lymphocytes, susceptibility of the recipient's immune system to their engraftment and degree of immunological (HLA) disparity between donor and patient. The minimum number of transfused lymphocytes necessary to provoke a GvHD reaction is unknown and may vary by clinical setting.

Implicated components typically reported include whole blood and red cells. This risk is higher in fresh units as ≤10 days oldwith no cases implicating components stored for > 2 weeks. Leucocyte depletion has been considered as a protective intervention.However, TA-GvHD continues to be reported within the era of leucocyte depletion.Features of TA-GvHD include rash, fever, elevated liver enzymes, pancytopenia, diarrhea, bone marrow aplasiaor hypocellularityand hepatomegaly. These occur 1– 6 weeks after transfusion, with the median time from transfusion to first symptom being 11 days. Overall survival rate is reported to be 8.4%.Diagnosis is usually made by biopsy of skin, gut, or liver. The presence of donor cells can be demonstrated by DNA amplification in peripheral bloodor short tandem repeat analysis using peripheral blood and skin biopsies from affected and non-affected sites in the patient, and peripheral blood samples from the implicated donors.Fluorescence in situ hybridisation (FISH) can be used for the diagnosis of TA- GvHD in sex mis-matched cases with rapid turnaround of results.

TA-GvHD is a rare condition with significant mortality and no randomized control trials available. It should be noted that the literature is scant, and levels of evidence are low. In addition, local practices, and processes for the development of blood components vary among countries. For these reasons there is considerable variation in recommendations between different national guidelines.

components at a given absorbed dose. For all at-risk patients, all red cell, platelet, and granulocyte components should be irradiated, except cryopreserved red cells after deglycerolisation. It is not necessary to irradiate fresh frozen plasma, cryoprecipitate, or fractionated plasma

2. Status of irradiation in Oman:

Irradiated blood components are supplied by department of blood bank services to all institutions within ministry of health. These specialized components are provided on a patient named basis only. Irradiation is by gamma rays.

3. Effects of irradiation on the blood components:

Leakage of $K^+$from red cells, hence the shelf life of red cell units is shortened, and risk of hyperkalaemia increases as the unit ages during storage.

4. Blood prescription/authorisation and administration issues:

- It is the responsibility of the clinical team involved with the patient's care to identify patients at risk of TA-GvHD, to inform the transfusion laboratory, and to request and prescribecellular blood components as irradiated.
- This will facilitate appropriate bedside checks and ensure patients' specific requirements are met.
- For patients under shared care, clinical areas and transfusion laboratories must ensure adequate communication about the requirement for irradiated components.
- Patients requiring irradiated components should be informed of the need and the reasons for this.

5. Approved clinical indications for use of Irradiated blood componentsatthe Royal Hospital:

i. **Bone marrow transplant:**

- All recipients (adult and paediatric) of allogeneic HSCT should receive irradiated blood components from the time of initiation of conditioning chemo/ radiotherapy.
- Irradiated components should be continued until all of the following criteria are met:

1. >6 months have elapsed since the transplant date
2. The lymphocyte count is >1.0 X109/l
3. The patient is free of active chronic GvHD
4. The patient is off all immunosuppression

- Treatment with irradiated blood components should continue indefinitely if this is required based on transplant conditioning, underlying disease or previous treatment
- Patients (adult and paediatric) undergoing bone marrow or peripheral blood stem cell collections for future autologous re-infusion should receive irradiated cellular blood components for 7 days prior to and during the bone marrow/stem cell harvest to prevent the collection of viable allogeneic T lymphocytes, which can potentially withstand cryopreservation.

- All patients undergoing autologous stem cell transplant (ASCT) irrespective of underlying diagnosis or indication for this treatment should receive irradiated cellular blood components from initiation of conditioning chemo/radiotherapy until 3 months post-transplant (6 months if total body irradiation was used in conditioning) unless conditioning, disease or previous treatment determine indefinite duration

i. **Donations from family members (first and second degree) and HLA-selected donors** previously frozen) have all been implicated in TA-GvHDeven if the patient is immunocompetent.

i. **All granulocytes concentrates** should be irradiated before issue. They should be transfused with minimum delay.

i. **Hodgkin disease**

All adults and children with HL at any stage of the disease should have irradiated red cells and platelets indefinitely.

i. **Congenital immune deficienciesin infants and children**

- All severe congenital T-lymphocyte immunodeficiency syndromes with significant qualitativeor quantitative T-lymphocyte deficiency should be considered as indications for irradiation of cellular blood components. it is reasonable to give irradiated cellular blood components for patients with suspected congenital HLH and lymphopenia, until T-cell immunodeficiency has been excluded.
- Once a diagnosis of severe T-lymphocyte immunodeficiency has been suspected, irradiated components should be given while further diagnostic tests are being undertaken. A clinical immunologist should be consulted for advice in cases where there is uncertainty.
- There is no indication for irradiation of cellular blood components for infants or children with temporary defects of T-lymphocyte function as the result of a viral infection.
- There is also no indication for irradiaion of cellular blood components for adults or children who are HIV-antibody positive or who have acquired immune deficiency syndrome (AIDS).
- Neonates and infants with suspected immunodeficiency syndromes should undergo T-lymphocyte enumeration prior to cardiac surgery wherever possible. If the T-lymphocyte count is >400 cells/l, of which 30% are naive T lymphocytes, there is no need to irradiate red cells or platelets. If it is not possible to undertake T-cell investigations prior to surgery, irradiated cellular blood components should be given until immunological investigations have been undertaken.
- To date, there have been no reports of TA- GvHD occurring in patients with isolated defects of humoral immunity.
- Adults, and children aged > 2 years without a significant history of infections, referred for elective cardiac surgery for problems associated with DiGeorge syndrome, such as aortic arch anomalies and pulmonary artery stenosis, or in whom DiGeorge anomaly is suspected, do not need to receive irradiated cellular blood components, unless there is a significant history consistent with severe T-lymphocyte-associated immunodeficiency.

i. **Aplastic anaemia** patients less than 40 years of age who are potential bone marrow transplant (BMT) candidatesundergoing treatment with ATG or alemtuzumab should receive irradiated blood components.

i. **Intrauterine transfusions (IUT) and neonatal red cell exchange transfusion (EBT)**

The risk of TA-GvHD in the fetus and neonate, especially if preterm, is thought to be higher than in older recipients. Contributing factors are the immunological immaturity,the use of fresher bloodand transfusion of relatively large volumes in situations such asEBT. Donor lymphocytes have been found in the neonatal circulation up to 6–8 weeks after EBTand cases of TA-GvHD have been described in neonates and young infants.This recommendation was based on the presumption of transfusion-induced tolerance or immune suppression following IUT and the knowledge of cases of TA-GvHD in the situation of IUT followed by EBT, rather than on specific evidence.

Routine irradiation of red cells and platelets for transfusion to preterm or term infants (other than for EBT) is not required unless there has been a previous IUT, in which case irradiated components should be administered until 6 months after the expected delivery date (40 weeks gestation).

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Written By: Dr Halima Al Balushi
Checked By: Dr Muna Al Riyami
Authorized by: Sabria Nasser Khalfan Al Hashami

Review Details

	Review BY	Review Date
1	Dr Sabriya Al Hashami	24/01/2024

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